Embedded System HW Design Week1

- Overview of embedded systems, architecture, and STM32 platform

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What is an Embedded System?

- **Computing system with a dedicated purpose**
  - Used in automotive, medical, robotics, consumer devices
  - Optimized for performance, power, and reliability
- **Code runs on application specific Hardware**
- **Typically have timing constraints**
  - Hard
  - Soft
- **Typical Hardware found in Embedded Systems**
  - Microprocessor
  - Memory
  - I/O ports
- **Typical Hardware NOT found in Embedded Systems**
  - Keyboard
  - Monitor
  - Disk Drive

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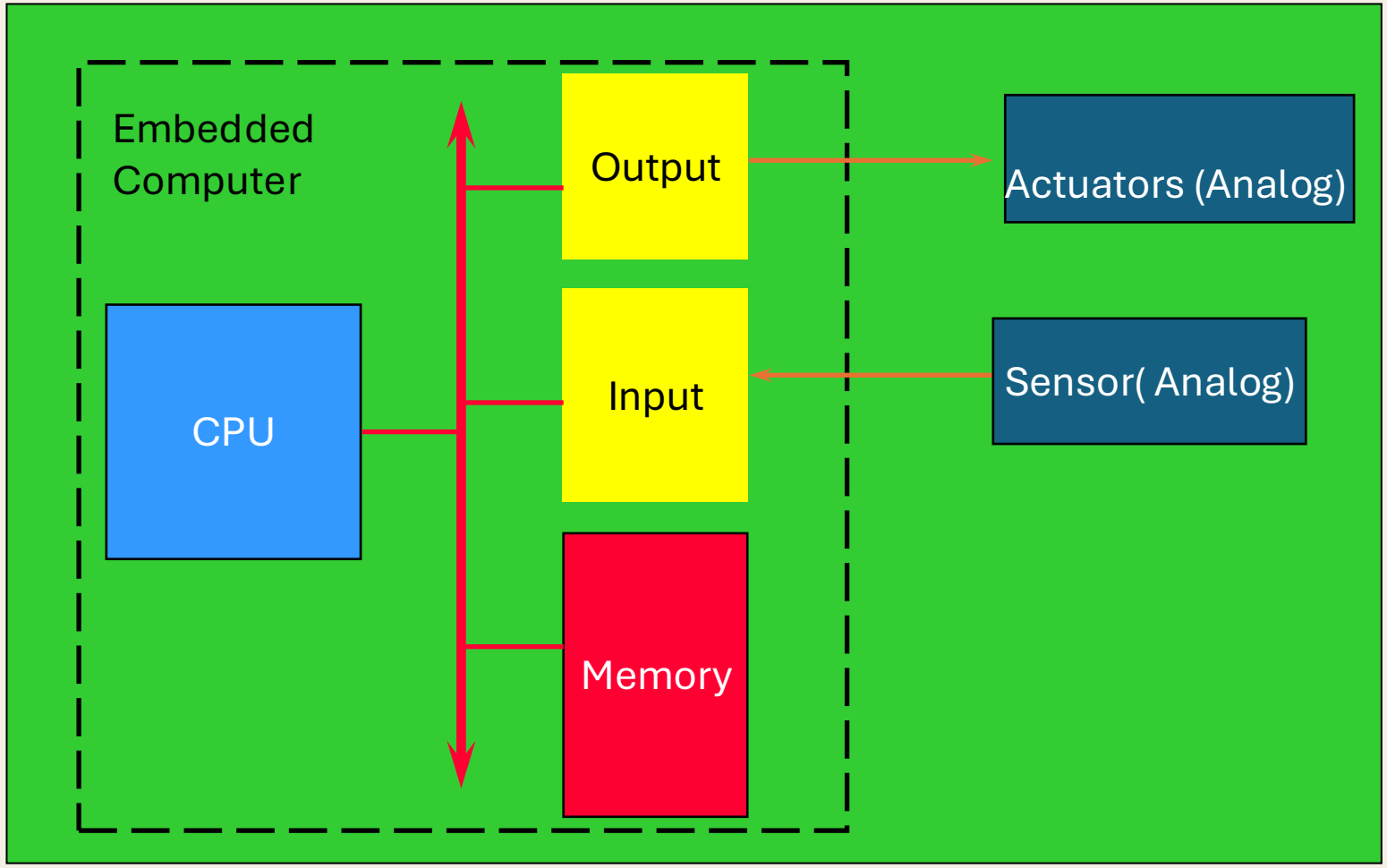
Embedded System



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Real-World Examples

- Smart thermostats
- Wearables
- Home security sensors
- Industrial controllers
- IoT environmental monitors

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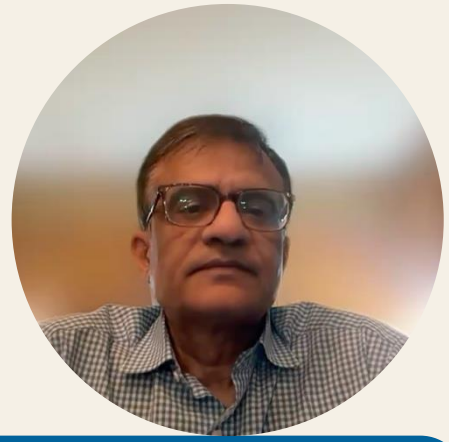
Core Components of an Embedded System

- Microcontroller (MCU)
- Memory (Flash, SRAM)
- Sensors and actuators
- Power subsystem
- Communication interfaces (UART, SPI, I2C, RF)



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Characteristics of embedded systems

- Sophisticated functionality.
- Real-time operation.
- Low manufacturing cost.
- Low power.
- Designed to tight deadlines by small teams.

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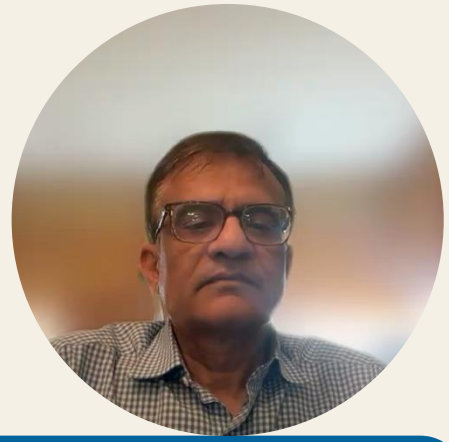
Functional complexity

- Often have to run sophisticated algorithms or multiple algorithms.
  - Cell phone, laser printer.
- Often provide sophisticated user interfaces.



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Real-time operation

- Must finish operations by deadlines.
  - Hard real time: missing deadline causes failure.
  - Soft real time: missing deadline results in degraded performance.
- Many systems are multi-rate: must handle operations at widely varying rates.

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Non-functional requirements

- Many embedded systems are mass-market items that must have low manufacturing costs.
  - Limited memory, microprocessor power, etc.
- Power consumption is critical in battery-powered devices.
  - Excessive power consumption increases system cost even in wall-powered devices.

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Challenges in embedded system design

- How much hardware do we need?
  - How big is the CPU? Memory?
- How do we meet our deadlines?
  - Faster hardware or cleverer software?
- How do we minimize power?
  - Turn off unnecessary logic? Reduce memory accesses?

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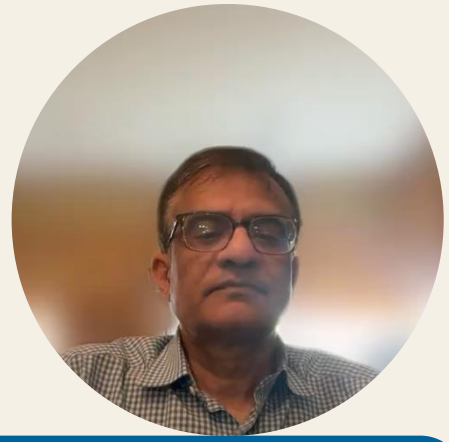
Internal Communication Flow

- CPU connects to peripherals through internal buses
  - AHB, APB, DMA subsystems
  - Impact on system performance and timing



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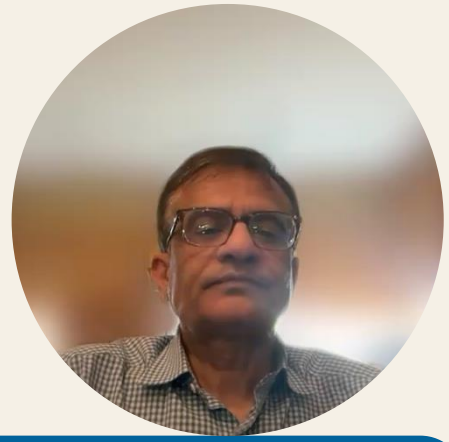
Why Architecture Matters

- **Strong architecture enables you to:**
- Evaluate design tradeoffs early
- Understand cost and component constraints
- Predict performance and timing limitations
- Design clean, well-defined interfaces

- **Poor architecture often results in:**
- Mismatched or incompatible components
- Signal integrity and noise issues
- Power and reliability problems
- Long debug cycles and costly redesigns



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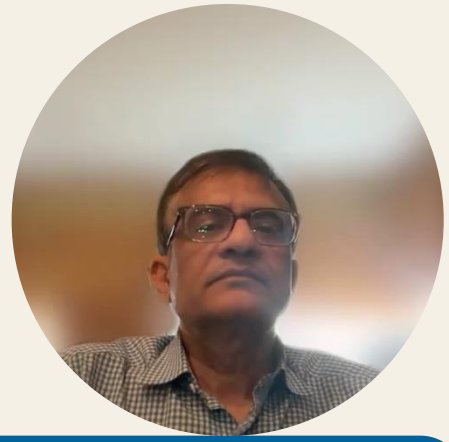
COTS(Commercial Off-The-Shelf) Modules

- **Why COTS Modules Are Used**
- Reduce development time and engineering risk
- Avoid designing everything from scratch
- Enable faster prototyping and time-to-market

- **Common COTS Modules**
- Wireless connectivity (Wi-Fi, BLE, LoRa)
- GPS receivers
- IMUs (accelerometer / gyroscope)
- Motor drivers
- Power regulation modules

- **Key Considerations When Selecting COTS**
- Electrical compatibility (voltage, current, signals)
- Communication interfaces (I^2C, SPI, UART, etc.)
- Mechanical constraints (size, connectors, mounting)
- Software and driver support



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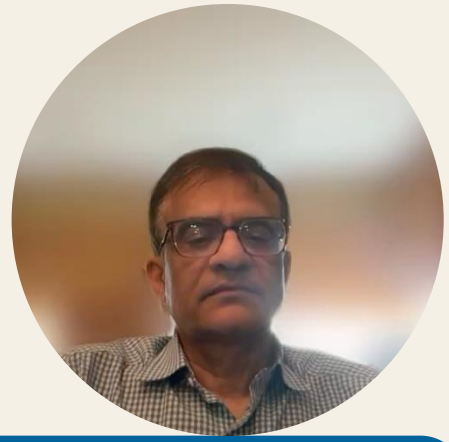
Introduction to STM32 IoT Discovery Kit

- Board used in this course: B-L475E-IOT01A
  - Features: Cortex-M4, sensors, BLE, WiFi, NFC
  - Ideal for learning embedded integration



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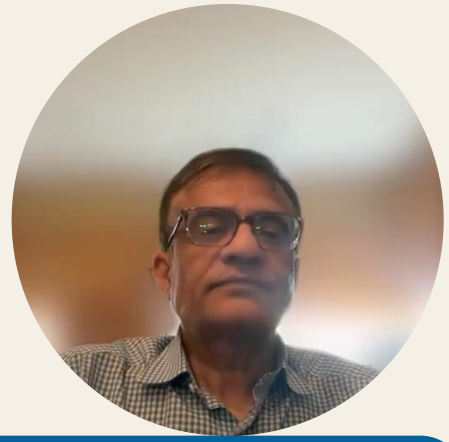
Week 1 Lab Overview

- Install STM32CubeIDE and CubeMX
  - Run LED blink + UART print

- Construct a block diagram
  for a real-world embedded system



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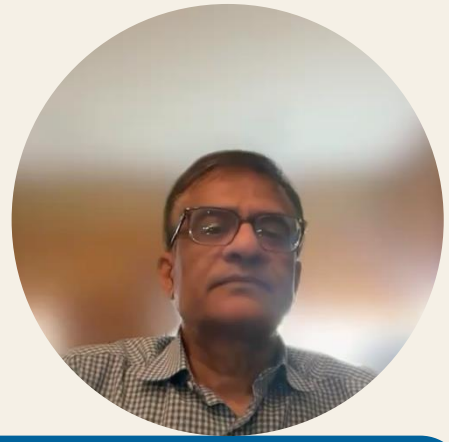

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